



Cornell University
Master Beekeeper Program

Field Exam

Below is the rubric for the Cornell University Master Beekeeper Field Exam (part 1 for virtual students). Students must conduct a colony inspection according to the methods taught in class. Students do not need to go through every frame in the hive during the exam.

The field exam (in person or recorded) can be a **maximum of 30 minutes** in length.

Criteria	Total possible	Student's score
Communication and Leadership		
The student leads the entire colony inspection, taking on the role as an expert teacher/mentor. The evaluator is not left with questions as they watch, such as “What are you doing now?” or “What do you see?” or “What does that mean?” because the student explains their steps, observations, and interpretations as they go along. The student's choice of inspection conditions is appropriate to this role.	30	
Evaluation of the apiary		
The student mentions two accurate “pros” of their apiary set up.	2	
The student mentions two accurate “cons” of their apiary and identifies potential impacts to the apiary/colony.	2	
Exterior Inspection		
Before opening the hive, the student makes three accurate observations about the bees and equipment.	3	
The student accurately describes what information these observations give to the beekeeper.	3	
Before opening the colony, the student can describe what he or she would expect to see inside the colony at this time of year.	5	

Hive opening and handling		
The student uses appropriate smoke when opening the colony.	1	
The student's method of opening the hive is conducted in an appropriate manner.	3	
The student handles bees and equipment carefully and mindfully. The student avoids squishing bees and uses the hive tool appropriately throughout the inspection.	5	
The student confidently and carefully pulls and handles frames from the hive without excessive damage to comb or bees.	2	
If there are supers present, the student is able to identify new wax/honey and evaluate whether it is time to put another super on. If it is not time, the student is able to describe when it is time to add a super.	2	
The student can assess whether there is a nectar flow occurring.	1	
The student places frames back in the same place and orientation in which they were removed.	1	
The student only leaves one frame (or two at maximum) out of the hive at a time.	1	
The student uses appropriate smoke throughout the colony inspection and remains in complete control of the colony throughout the inspection (i.e., the bees do not escalate into highly defensive behavior).	5	
Inspecting adults		
The student accurately counts the number of frames of bees (seams of bees) in a hive body, and comments on the population of bees in the colony. He or she interprets the population for the time of year accurately.	5	
The student identifies if the colony is experiencing a population increase or decrease.	2	
During the inspection, the student accurately identifies the different castes of bees. If a caste is not seen, the student accurately describes the characteristics of that caste at the end of the inspection.	30	
If the caste was obviously visible during the inspection, no points are given if missed by the student.		

The student examines the adult bees for potential health issues, and determines if these issues are present in the colony.	6	
The student comments on whether this colony was overwintered or if it was a nuc, package, or split of this year. (Not applicable for in-person exams.)	1	
The student is mindful of the potential for the queen to be on frames (checking for her before putting them on the ground, holding the frame over the hive initially, etc.).	1	
Inspecting brood		
The student describes and interprets the brood pattern of the colony. If the pattern is spotty, the student lists valid reasons why.	5	
The student can identify different developmental stages of brood.	1	
The student comments on the amount of brood in the colony. He or she interprets the amount accurately for the time of year.	3	
The student can assess the queen status of the colony (queenright/queenless/drone layer, etc).	3	
The student examines the brood for potential health issues, and determines if these issues are present in the colony.	15	
The student correctly describes how a beekeeper can proactively maintain brood health in the apiary.	3	
The student can accurately determine whether or not there are eggs. Note: virtual exams must show the frame to document their observations.	30	
The student accurately identifies (or describes) different kinds of queen cups, and their purposes. If none are seen, the student accurately describes the characteristics and purposes of each.	4	
The student is able to describe what should be done to manage swarming if charged swarm cells existed.	2	
The student is able to evaluate the age of the comb	1	
Evaluating Nutrition		
The student can describe where they would expect to see pollen and honey in the colony.	1	

The student can accurately identify pollen and honey and state what nutrients they provide bees.	4	
The student is able to evaluate whether a colony has enough pollen by examining royal jelly levels.	1	
The student comments on the amount of food in the colony. He or she interprets the amount accurately for the time of year.	2	
After closing up the hive		
The student's method of closing the hive is conducted in an appropriate manner.	3	
The student can knowledgably discuss <i>Varroa</i> management practices according to the IPM principles taught in class.	30	
The student describes how often a colony should be inspected during various times of the year.	2	
The student describes whether the colony is healthy overall, and explain why they came to that conclusion.	4	
The student describes whether the colony requires any immediate management (fixing queen issues, addressing health issues, provide more room etc.), and what the goals for the next inspection should be.	5	

There are four circumstances where a student automatically fails an exam, despite their scores on the above criteria:

1. The student is unable to control the colony's defensiveness and the bees' behavior becomes dangerous OR the student is oblivious to the colony escalating into highly defensive behavior.
2. The student is observed accidentally killing a queen.
3. The student is observed using illegal/off-label/unregistered treatments or recommends using such treatments during their exam.
4. Another person present during the exam comments or points out criteria we are grading. For example, the videographer proclaims, "there's the queen!". (You may have someone filming or helping you lift heavy hive bodies, but they cannot participate in the exam.)